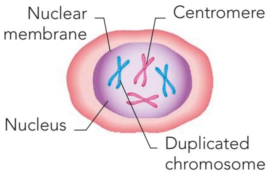
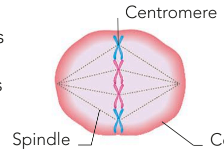
MAKING NEW BODY CELLS

Cell division (mitosis) produces new cells for growth, maintenance, and repair. First, all the DNA replicates and the chromosomes are duplicated. These double-chromosomes form a line and then migrate away from each other as the cell splits in two.

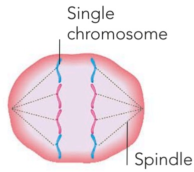
1 PREPARATION DNA replicates and forms double-chromosomes. The nuclear membrane breaks down.



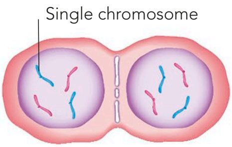
2 ALIGNMENT A spindle fibre holds the centromere of each double-chromosome as it lines up in the middle of the cell.



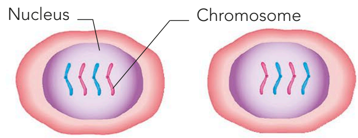
3 SEPARATION Each centromere splits so that single chromosomes move to each end of the cell.



4 SPLITTING As the spindle disappears, nuclear membranes form around the two groups of chromosomes.



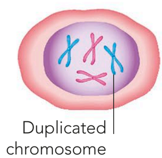
5 OFFSPRING The cytoplasm divides and the cell splits into two. Both the new cells have 23 pairs of chromosomes. Only two pairs are shown here for clarity.



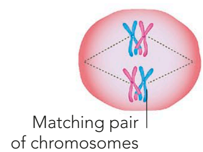
MAKING SEX CELLS

Sperm and egg cells divide by meiosis, into four sex cells (eggs or sperm) that have only one member of a chromosome pair. At fertilization, when egg and sperm unite, the full set (23 pairs) is restored and all subsequent cell divisions are by mitosis.

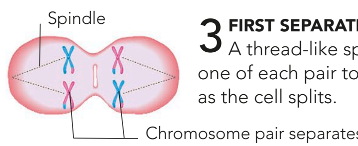
1 PREPARATION DNA strands replicate and coil up in the nucleus, forming X-shaped double-chromosomes.



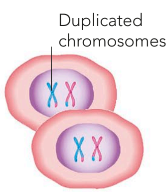
2 PAIRING The matching (homologous) pairs align, make contact, and exchange genetic material.



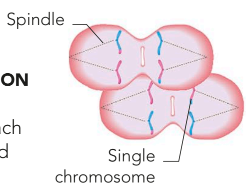
3 FIRST SEPARATION A thread-like spindle pulls one of each pair to each end as the cell splits.



4 TWO OFFSPRING Each cell has one double-chromosome of each pair, as a random choice during separation.



5 SECOND SEPARATION The double-chromosome splits, each half moving to one end of the dividing cell.



6 FOUR OFFSPRING The four sex cells differ from each other and the parent cell in their genetic composition.